INTRODUCTION

Among the pioneering works involving biological oxidation, Antoine Lavoisier, who is known as the father of modern chemistry, in 1789 demonstrated that aerobic systems consume oxygen and generate carbon dioxide to obtain enormous amount of energy as compared to anaerobic systems.

Aerobic organisms achieve it through oxidation of glucose and fatty acids. Oxygen is a highly reactive molecule and therefore its use is tightly regulated. Inappropriate use of oxygen accumulates toxic products and therefore aerobic organisms possess multiple mechanisms to protect themselves against the deleterious effects of oxygen. Antioxidant enzymes such as superoxide dismutase and catalase protect cells against oxidative damage. However, despite this precision, oxygen metabolites sometimes cause damage to living cells and collapse the antioxidant mechanisms. Life threatening disorders such as cancer, neuro-degeneration, diabetes are known to be caused in part due to oxygen metabolites.

The complexity of aerobic cells is due to the presence of thousands of small molecules organized as molecular machines. These molecular machines are proteins that use chemical bond energy of nucleotides to attain energy. The structure of the machine is designed to bind Adenosine tri phosphate (ATP) in a specific orientation. Aerobic eukaryotic cells possess variety of molecular machines. Higher animals are absolutely dependent upon a steady supply of oxygen for respiration, the process through which cells derive energy in the form of ATP.

Otto Warburg in his studies demonstrated that biological oxidations are catalyzed by enzymes designated as oxygenases. Many drugs and toxic metabolites are metabolized by a specialized enzyme class known as mixed function oxidases (cytochrome P450 system).

Harvesting of energy from glucose has three stages (1) Glycolysis (breaks down glucose into two molecules of pyruvate), (2) The citric acid cycle (completes the breakdown of glucose) and (3) Oxidative phosphorylation (accounts for most of the ATP synthesis).

8.1 REDOX REACTION

Biological electron transport is a series of coupled oxidation and reduction reactions. The oxidation of a molecule involves the loss of electrons and reduction involves gain of electrons.

$$\text{Reduced compound } (e^- \text{ donor}) \rightleftharpoons \text{Oxidized compound} + e^-$$

$$\text{Eg.} \qquad Fe^{3+} \rightleftharpoons Fe^{2+} + e^-$$

A complete redox reaction must have one reactant, an electron acceptor, which gets reduced by gaining electrons. The electron donor is called as reductant which becomes oxidized while transfering electrons to the other substrate which is called as the oxidant.

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